



Machine Learning

Quiz - 3 (15 Marks) (Duration 60 min*)



Beginners (8 marks)

- What is the objective of the E-step (Expectation) in the EM algorithm for GMMs?
Cluster Assignment
- Are posterior probabilities produced by a Naive Bayes classifier necessarily well-calibrated? Briefly explain.
Not necessarily. Naive Bayes often gives poorly calibrated probabilities because of the conditional-independence assumption
- Under what condition does the GMM likelihood function approach infinity?
The log-likelihood of a GMM becomes unbounded above when a single Gaussian component *collapses* onto one data point x_i . If a component mean μ_k equals x_i and the covariance determinant $|\Sigma_k| \rightarrow 0$, then

$$\mathcal{N}(x_i; \mu_k, \Sigma_k) = \frac{1}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \exp\left[-\frac{1}{2}(x_i - \mu_k)^\top \Sigma_k^{-1} (x_i - \mu_k)\right] \rightarrow \infty.$$

Thus, for that data point,

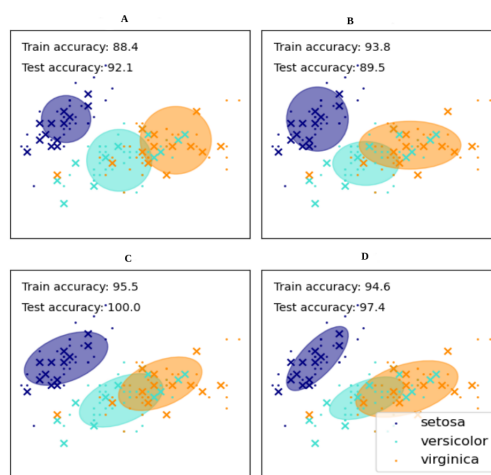
$$\log(\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k)) \approx -\frac{1}{2} \log |\Sigma_k| + \text{const} \rightarrow +\infty.$$

Hence, $\mathcal{L}(\theta) \rightarrow +\infty$ when any Σ_k collapses to zero around a sample.

- Under what specific assumptions would a GMM's clustering result be identical to a K-Means clustering result?
1) All Gaussian components share the same spherical covariance matrix. 2) Responsibilities γ are hard 0/1
- A GMM is known to be identifiable up to permutation of components under which of the following regularity conditions:
A. K is not fixed B. Any two components are identical C. Covariances are singular D. All of the above
E. None of the above D

Intermediate

- Given these plots, write a combination of same/different and full/diagonal covariance types for each plot, with justification.
4 marks



- A - diagonal-different
- B - diagonal-different
- C - full-same
- D - full-different

Partial credit for answers (8 components, total 2.5 marks):

Correct	Marks
1	0.3
2	0.6
3	0.9
4	1.3
5	1.6
6	1.9
7	2.3
8	2.5

Justification definitions (1.5 marks):

- Diagonal: axis-aligned, no rotation/tilt
- Full: rotated/tilted, not axis-aligned
- Same: same size, shape, orientation
- Different: differs in size or shape

Partial credit for justification (4 components, total 1.5 marks):

Correct	Marks
1	0.4
2	0.8
3	1.2
4	1.5

7. Comment on the Naive Bayes property of robustness to isolated noisy points in general?
Depends on the distribution, e.g., Gaussian NB is robust while Bernoulli-NB is not.

2 marks

Advanced (4 marks)

8. Consider the log-likelihood of a Gaussian Mixture Model (GMM)

$$\log p(X | \theta) = \sum_{n=1}^N \log \left(\sum_{k=1}^K \pi_k \mathcal{N}(x_n; \mu_k, \Sigma_k) \right),$$

where π_k are mixture weights and $\mathcal{N}(x_n; \mu_k, \Sigma_k)$ is the Gaussian density.

Let q_{nk} be any distribution over the latent assignment variables z_{nk} , satisfying $q_{nk} \geq 0$ and $\sum_k q_{nk} = 1$.

(a) Using Jensen's inequality, derive a lower bound on the log-likelihood of the form

$$\log p(X | \theta) \geq \sum_{n=1}^N \sum_{k=1}^K q_{nk} [\log \pi_k + \log \mathcal{N}(x_n; \mu_k, \Sigma_k) - \log q_{nk}].$$

(b) Show clearly:

1. how the log of a sum is rewritten as an expectation w.r.t. q_{nk} ,
2. where Jensen's inequality is applied,
3. and why the inequality becomes equality when

$$q_{nk} = P(z_{nk} = 1 | x_n, \theta).$$

We start from the per-sample term in the log-likelihood of a Gaussian mixture:

$$\log p(x_n | \theta) = \log \left(\sum_{k=1}^K \pi_k \mathcal{N}(x_n; \mu_k, \Sigma_k) \right),$$

where $\{\pi_k\}$ are mixture weights and $\mathcal{N}(\cdot; \mu_k, \Sigma_k)$ denotes a Gaussian density.

Step 1: introduce an arbitrary auxiliary distribution $q_n = (q_{n1}, \dots, q_{nK})$. Let $q_{nk} \geq 0$ and $\sum_{k=1}^K q_{nk} = 1$. Using the identity

$$\sum_k \pi_k p_k(x_n) = \sum_k q_{nk} \frac{\pi_k p_k(x_n)}{q_{nk}},$$

we rewrite the log-sum as

$$\log p(x_n | \theta) = \log \left(\sum_{k=1}^K q_{nk} \frac{\pi_k \mathcal{N}(x_n; \mu_k, \Sigma_k)}{q_{nk}} \right).$$

Reaching the aforementioned checkpoint / equivalent expression earns you 2 marks

Step 2: apply Jensen's inequality. The logarithm is concave, so for any probability vector q_n ,

$$\log(\mathbb{E}_{q_n}[A_k]) \geq \mathbb{E}_{q_n}[\log A_k].$$

Set $A_k = \frac{\pi_k \mathcal{N}(x_n; \mu_k, \Sigma_k)}{q_{nk}}$. Applying Jensen gives

$$\log p(x_n | \theta) \geq \sum_{k=1}^K q_{nk} \log \left(\frac{\pi_k \mathcal{N}(x_n; \mu_k, \Sigma_k)}{q_{nk}} \right).$$

Step 3: expand the right-hand side.

$$\begin{aligned} \log p(x_n | \theta) &\geq \sum_{k=1}^K q_{nk} \left[\log \pi_k + \log \mathcal{N}(x_n; \mu_k, \Sigma_k) - \log q_{nk} \right] \\ &= \sum_{k=1}^K q_{nk} \log \pi_k + \sum_{k=1}^K q_{nk} \log \mathcal{N}(x_n; \mu_k, \Sigma_k) - \sum_{k=1}^K q_{nk} \log q_{nk}. \end{aligned}$$

Step 4: sum over all data points to obtain the global lower bound. Summing the per-sample bounds for $n = 1, \dots, N$ gives

$$\log p(X | \theta) = \sum_{n=1}^N \log p(x_n | \theta) \geq \sum_{n=1}^N \sum_{k=1}^K q_{nk} \left[\log \pi_k + \log \mathcal{N}(x_n; \mu_k, \Sigma_k) - \log q_{nk} \right].$$

Reaching the aforementioned checkpoint / equivalent expression earns you 1 mark

Define the RHS as the variational lower bound $\mathcal{L}(q, \theta)$.

Step 5: tightness condition (when inequality is equality). The Jensen inequality becomes an equality iff the argument of the log is constant q_n -almost surely, which here implies

$$\frac{\pi_k \mathcal{N}(x_n; \mu_k, \Sigma_k)}{q_{nk}} = C_n \quad (\text{independent of } k).$$

Re-arranging yields

$$q_{nk} \propto \pi_k \mathcal{N}(x_n; \mu_k, \Sigma_k),$$

and normalizing over k gives exactly the posterior responsibility

$$q_{nk} = P(z_{nk} = 1 \mid x_n, \theta) = \frac{\pi_k \mathcal{N}(x_n; \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x_n; \mu_j, \Sigma_j)}.$$

Thus choosing q_{nk} equal to the posterior makes the Jensen bound tight; this is the E-step of EM.

Conclusion: The derived lower bound is

$$\log p(X \mid \theta) \geq \mathcal{L}(q, \theta) = \sum_{n=1}^N \sum_{k=1}^K q_{nk} \left[\log \pi_k + \log \mathcal{N}(x_n; \mu_k, \Sigma_k) - \log q_{nk} \right],$$

and equality holds when $q_{nk} = P(z_{nk} = 1 \mid x_n, \theta)$.

Proving the equivalence under specified conditions and reasoning earns you 1 mark.